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Conformal Kaehler Euclidean submanifolds

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ABSTRACT

Let $f\colon M^{2n} \to \mathbb{R}^{2n+\ell}$, $n \geq 5$, be a conformal immersion into Euclidean space with codimension ℓ where M^{2n} is a Kaehler manifold of complex dimension n free of points where all sectional curvatures vanish. For codimension $\ell = 1$ or $\ell = 2$ we show that at least locally such a submanifold can always be obtained in a rather simple way, namely, from an isometric immersion of the Kaehler manifold M^{2n} into either $\mathbb{R}^{2n+1}$ or $\mathbb{R}^{2n+2}$, the latter being a class of submanifolds already extensively studied.

Throughout the paper $f\colon M^{2n} \to \mathbb{R}^{2n+\ell}$ denotes a *conformal Kaehler submanifold*, that is, (M^{2n}, J) is a connected Kaehler manifold of complex dimension $n \geq 2$ and f is a conformal immersion into Euclidean space with codimension ℓ. That the immersion is *conformal* means that there is a positive function $\lambda \in C^\infty(M)$ such that the metric induced by f relates to the original Kaehler metric by $\langle\, ,\,\rangle_f = \lambda^2 \langle\, ,\,\rangle_{M^{2n}}$. The immersion f is called a *real Kaehler submanifold* if $\lambda \equiv 1$. Our goal is to describe, up to a conformal congruence of the ambient space, the local situation of the conformal Kaehler submanifolds if ℓ is at most two. Recall that a pair of immersions $f, g\colon M^n \to \mathbb{R}^N$ are said to be *conformally congruent* if $g = \tau \circ f$ for some conformal (Moebius) transformation τ of $\mathbb{R}^N$.

In the following result by a *real Kaehler hypersurface* we mean a real Kaehler submanifold with codimension one. This class of submanifolds have been locally classified by Dajczer and Gromoll [4] by means of the Gauss parametrization in terms of a pseudoholomorphic surface in a sphere and a smooth function on the surface. Florit and Zheng [9] showed that metrically complete real Kaehler hypersurfaces are just cylinders over a surface in $\mathbb{R}^3$.

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Theorem 1. *Any conformal immersion $f\colon M^{2n} \to \mathbb{R}^{2n+1}$, $n \geq 4$, of a Kaehler manifold free of points where all sectional curvatures vanish is conformally congruent to a real Kaehler hypersurface.*

For codimension two, simple examples of conformal Kaehler submanifolds are obtained by composing a holomorphic hypersurface $M^{2n} \to \mathbb{C}^{n+1}$ or the extrinsic product of a pair of real Kaehler hypersurfaces with a conformal transformation of the ambient space. But there are many other examples of real Kaehler submanifolds that can be composed with a conformal ambient transformation; for instance see Dajczer and Gromoll [5] for the class of complex ruled submanifolds, including metrically complete ones, produced by means of a Weierstrass type representation. See also Dajczer and Florit [3] for the case of submanifolds of rank two.

Theorem 2. *Let $f\colon M^{2n} \to \mathbb{R}^{2n+2}$, $n \geq 5$, be a conformal Kaehler submanifold where M^{2n} is free of points where all sectional curvatures vanish. Then there is an open dense subset M_0 of M^{2n} such that along any connected component N^{2n} of M_0 one of the following holds:*

(i) *$f|_N$ is conformally congruent to a real Kaehler submanifold $g\colon N^{2n} \to \mathbb{R}^{2n+2}$.*
(ii) *$f|_N = h \circ g$ is a composition of a real Kaehler hypersurface $g\colon N^{2n} \to \mathbb{R}^{2n+1}$ and a conformal immersion $h\colon V \to \mathbb{R}^{2n+2}$ where $V \subset \mathbb{R}^{2n+1}$ is open and $g(N) \subset V$.*

Notice that h in part (ii) is a conformally flat hypersurface. The submanifolds in this class have been parametrically described by do Carmo, Dajczer and Mercuri [1].

1. Preliminaries

1.1. The isometric light-cone representative

The *light-cone* $\mathbb{V}^{m+1}$ of the standard flat Lorentzian space $\mathbb{L}^{m+2}$ is any one of the two connected components of the set of all light-like vectors $\{v \in \mathbb{L}^{m+2} : \langle v, v\rangle = 0,\ v \neq 0\}$ endowed with the degenerate metric inherited from $\mathbb{L}^{m+2}$. The Euclidean space $\mathbb{R}^m$ can be realized as an umbilic hypersurface of $\mathbb{V}^{m+1}$ as follows: Given light-like vectors $v, w \in \mathbb{L}^{m+2}$ such that $\langle v, w\rangle = 1$ and a linear isometry $C\colon \mathbb{R}^m \to \{v, w\}^\perp$, define $\Psi\colon \mathbb{R}^m \to \mathbb{V}^{m+1} \subset \mathbb{L}^{m+2}$ by

$$\Psi(x) = v + Cx - \frac{1}{2}\|x\|^2 w.$$

Then Ψ is an isometric embedding of $\mathbb{R}^m$ as an umbilical hypersurface in the light cone given as an intersection of $\mathbb{V}^{m+1}$ with an affine hyperplane, namely,

$$\Psi(\mathbb{R}^m) = \{y \in \mathbb{V}^{m+1} \colon \langle y, w\rangle = 1\}. \tag{1}$$

The normal bundle of Ψ is $N_\Psi\mathbb{R}^m = \operatorname{span}\{\Psi, w\}$ and its second fundamental form is

$$\alpha^\Psi(X, Y) = -\langle X, Y\rangle_{\mathbb{R}^m} w.$$

We observe that $\Psi(\mathbb{R}^m)$ is independent of the triple $\{v, w, C\}$ in the sense that different triples produce submanifolds congruent by an isometry of $\mathbb{L}^{m+2}$.

If $f\colon M^n \to \mathbb{R}^m$ is a conformal immersion with conformal factor $\lambda \in C^\infty(M)$, then the isometric immersion

$$F = \frac{1}{\lambda}\Psi \circ f\colon M^n \to \mathbb{V}^{m+1} \subset \mathbb{L}^{m+2},$$

is called the *isometric light-cone representative* of f. Differentiating $\langle F, F\rangle = 0$ gives that $F \in \Gamma(N_F M)$. Differentiating once more yields that the second fundamental form of F satisfies

$$\langle \alpha^F(X,Y), F\rangle = -\langle X, Y\rangle \tag{2}$$

for any tangent vector fields $X, Y \in \mathfrak{X}(M)$. Since $\Psi_* N_f M \subset N_F M$ then the normal bundle of F decomposes as $N_F M = \Psi_* N_f M \oplus L^2$ where L^2 is the Lorentzian plane subbundle in the normal bundle of F orthogonal to $\Psi_* N_f M$ and such that $F \in \Gamma(L^2)$. The full expression of the second fundamental form of F as well as additional information on the isometric light-cone representatives is given in [7] and [10].

Proposition 3. *Two conformal immersions $f, g\colon M^n \to \mathbb{R}^m$ are conformally congruent if and only if their isometric light-cone representatives $F, G\colon M^n \to \mathbb{V}^{m+1} \subset \mathbb{L}^{m+2}$ are isometrically congruent.*

Proof. See Proposition 9.18 in [7]. $\square$

Proposition 4. *Let $F\colon M^n \to \mathbb{V}^{n+p+1} \subset \mathbb{L}^{n+p+2}$ be an isometric immersion that carries a normal light-like vector field δ which is constant in $\mathbb{L}^{n+p+2}$ and satisfies $\langle F, \delta\rangle = 1$. Then there is an isometric immersion $f\colon M^n \to \mathbb{R}^{n+p}$ which has F as its isometric light-cone representative.*

Proof. Let $v \in \mathbb{L}^{n+p+2}$ a light-like vector such that $\langle v, \delta\rangle = 1$ and let $C\colon \mathbb{R}^{n+p} \to \{v, \delta\}^\perp$ be a linear isometry. Then let $\Psi\colon \mathbb{R}^{n+p} \to \mathbb{V}^{n+p+1} \subset \mathbb{L}^{n+p+2}$ be the isometric embedding given by $\Psi(x) = v + Cx - \frac{1}{2}\|x\|^2\delta$. Since we have from (1) that $F(M) \subset \Psi(\mathbb{R}^{n+p})$ the proof follows. $\square$

1.2. Flat bilinear forms

Let V^n and $W^{p,p}$ be real vector spaces of dimensions n and $2p$, respectively. We assume that the latter is endowed with an inner product of signature (p, p). This means that p is the dimension of the subspaces of maximal dimension where the inner product is either positive or negative definite. A vector subspace $L \subset W^{p,p}$ is called *degenerate* if $L \cap L^\perp \neq \{0\}$ and *nondegenerate* otherwise. That L is *isotropic* means that $L = L^\perp$.

A bilinear form $\beta\colon V^n \times V^n \to W^{p,p}$ (maybe not symmetric) is called *flat* if

$$\langle \beta(X,Y), \beta(Z,T)\rangle - \langle \beta(X,T), \beta(Z,Y)\rangle = 0$$

for all $X, Y, Z, T \in V^n$. It is said that β is *null* when

$$\langle \beta(X,Y), \beta(Z,T)\rangle = 0$$

for all $X, Y, Z, T \in V^n$. Thus null bilinear forms are trivially flat. We denote

$$\mathcal{S}(\beta) = \operatorname{span}\{\beta(X,Y)\colon X, Y \in V^n\}$$

and

$$\mathcal{N}(\beta) = \{Y \in V^n\colon \beta(X,Y) = 0 \text{ for all } X \in V^n\}.$$

Proposition 5. *Let V^n and U^p, $2p \leq n$ and $1 \leq p \leq 5$, be real vector spaces such that U^p carries an inner product of any signature and $J \in End(V)$ satisfies $J^2 = -I$. Then let $W^{p,p} = U^p \oplus U^p$ be endowed with the inner product of signature (p, p) given by*

$$\langle\langle(\xi_1,\xi_2),(\eta_1,\eta_2)\rangle\rangle = \langle\xi_1,\eta_1\rangle_{U^p} - \langle\xi_2,\eta_2\rangle_{U^p}. \tag{3}$$

Let $\alpha\colon V^n \times V^n \to U^p$ be a symmetric bilinear form and $\beta\colon V^n \times V^n \to U^p \oplus U^p$ the bilinear form given by

$$\beta(X,Y) = (\alpha(X,Y), \alpha(X,JY)). \tag{4}$$

If β is flat and the subspace $\mathcal{S}(\beta)$ is nondegenerate then $\dim \mathcal{N}(\beta) \geq n-2p$.

For the proof we use the following elementary fact.

Lemma 6. *Let V be a real vector space and let $J \in End(V)$ satisfy $J^2 = -I$. Assume that $X_1, JX_1, \ldots, X_{k-1}, JX_{k-1}, X_k \in V$ are linearly independent vectors. Then also $X_1, JX_1, \ldots, X_k, JX_k$ are linearly independent. In particular, the dimension of V is even.*

Proof. If $JX_k = \sum_{i=1}^{k} a_iX_i + \sum_{j=1}^{k-1} b_jJX_j$ where $0 \neq (a_1,\ldots,a_k,b_1,\ldots,b_{k-1})$ then

$$(1+a_k^2)X_k + \sum_{j=1}^{k-1}((a_ka_j - b_j)X_j + (a_kb_j + a_j)JX_j) = 0,$$

and this is a contradiction. $\square$

Proof of Proposition 5. Let $B_Z\colon V^n \to W^{p,p}$ for $Z \in V^n$ be given by $B_ZY = \beta(Z,Y)$ and set $\mathcal{U}(Z) = B_Z(V) \cap B_Z(V)^\perp \subset W^{p,p}$. We denote

$$r_o = \max\{\dim B_Z(V)\colon Z \in V^n\}$$

and

$$\tau = \min\{\dim \mathcal{U}(Z)\colon Z \in V^n \text{ and } \dim B_Z(V) = r_o\}.$$

A simple argument gives that

$$RE^o(\beta) = \{Z \in V^n\colon r_o = \dim B_Z(V) \text{ and } \tau = \dim \mathcal{U}(Z)\}$$

is an open dense subset of V^n, for instance, see the proof of Lemma 2.1 in [6].

We have that $\tau \leq p-1$. In fact, if otherwise $\mathcal{U}(X) = B_X(V)$ if $X \in RE^o(\beta)$, and then the density of $RE^o(\beta)$ gives that

$$\langle\langle\beta(Z,Y), \beta(Z,T)\rangle\rangle = 0$$

for any $Z,Y,T \in V^n$. Now flatness yields

$$0 = \langle\langle\beta(Z+R,Y), \beta(Z+R,T)\rangle\rangle = 2\langle\langle\beta(Z,Y), \beta(R,T)\rangle\rangle$$

for any $Z,Y,T,R \in V^n$, contradicting that the subspace $\mathcal{S}(\beta)$ is nondegenerate.

Fix $X \in RE^o(\beta)$ and denote $N(X) = \ker B_X$. Proposition 4.6 in [7] gives that

$$\text{span}\,\{\beta(Y,Z)\colon Y \in V^n \text{ and } Z \in N(X)\} \subset \mathcal{U}(X). \tag{5}$$

Thus if $\tau = 0$ then $N(X) = \mathcal{N}(\beta)$ and hence $\dim \mathcal{N}(\beta) \geq n - 2p$ holds. Therefore, in the sequel we assume that $1 \leq \tau \leq p - 1 \leq 4$.

We have that $(\eta, \bar{\eta}) \in \mathcal{U}(X)$ if and only if $(\bar{\eta}, -\eta) \in \mathcal{U}(X)$. In fact, if $(\eta, \bar{\eta}) = B_X Z$ then $(\bar{\eta}, -\eta) = B_X JZ$. Moreover, since we have

$$0 = \langle\langle B_X Y, (\eta, \bar{\eta}) \rangle\rangle = \langle \alpha(X, Y), \eta \rangle - \langle \alpha(X, JY), \bar{\eta} \rangle$$

for any $Y \in V^n$, then also

$$\langle\langle B_X Y, (\bar{\eta}, -\eta) \rangle\rangle = \langle \alpha(X, Y), \bar{\eta} \rangle + \langle \alpha(X, JY), \eta \rangle = 0$$

for any $Y \in V^n$. Now Lemma 6 applied to $J \in \mathrm{End}(\mathcal{U}(X))$ defined by $J(\eta, \bar{\eta}) = (\bar{\eta}, -\eta)$ gives $\tau = 2s$ and

$$\mathcal{U}(X) = \mathrm{span}\,\{(\eta_1, \bar{\eta}_1), (\bar{\eta}_1, -\eta_1), \ldots, (\eta_s, \bar{\eta}_s), (\bar{\eta}_s, -\eta_s)\}. \tag{6}$$

By Corollary 4.3 in [7] there is the decomposition

$$U^p \oplus U^p = \mathcal{U}^\tau(X) \oplus \hat{\mathcal{U}}^\tau(X) \oplus \mathcal{V}^{p-\tau, p-\tau} \tag{7}$$

such that $\hat{\mathcal{U}}^\tau(X)$ is isotropic, the subspace $\mathcal{V} = (\mathcal{U}^\tau(X) \oplus \hat{\mathcal{U}}^\tau(X))^\perp$ is nondegenerate and $B_X(V) \subset \mathcal{U}(X) \oplus \mathcal{V}$. For later use notice that

$$\dim N(X) \geq n - 2p + \tau. \tag{8}$$

Let $\hat{\beta}$ denote the $\hat{\mathcal{U}}(X)$-component of β. That the subspace $\mathcal{S}(\beta)$ is nondegenerate gives that $\mathcal{S}(\hat{\beta}) = \hat{\mathcal{U}}(X)$. Setting $\hat{B}_Y Z = \hat{\beta}(Y, Z)$ we have from (6) and (7) that

$$\langle\langle \hat{B}_Y Z, (\eta_j, \bar{\eta}_j) \rangle\rangle = \langle\langle B_Y Z, (\eta_j, \bar{\eta}_j) \rangle\rangle = \langle \alpha(Y, Z), \eta_j \rangle - \langle \alpha(Y, JZ), \bar{\eta}_j \rangle$$

for any $Y, Z \in V^n$. Then

$$\begin{cases} \langle\langle \hat{B}_Y Z, (\eta_j, \bar{\eta}_j) \rangle\rangle = -\langle\langle \hat{B}_Y JZ, (\bar{\eta}_j, -\eta_j) \rangle\rangle, \ 1 \leq j \leq s, \\ \langle\langle \hat{B}_Y Z, (\bar{\eta}_j, -\eta_j) \rangle\rangle = \langle\langle \hat{B}_Y JZ, (\eta_j, \bar{\eta}_j) \rangle\rangle, \ 1 \leq j \leq s, \end{cases} \tag{9}$$

for any $Y, Z \in V^n$. It follows from (6) and (9) that $S \in End(\hat{\mathcal{U}}(X))$ given by $S\hat{B}_Y Z = \hat{B}_Y JZ$ is well defined. Since $S^2 = -I$ then Lemma 6 gives that

$$\kappa = \max\{\dim \hat{B}_Z(V) \colon Z \in V^n\}$$

is even and thus $2 \leq \kappa \leq \tau \leq 4$.

A simple argument shows that

$$RE(\hat{\beta}) = \{Y \in V^n \colon \ \dim \hat{B}_Y(V) = \kappa\}$$

is an open dense subset of V^n; see Proposition 4.4 in [7]. We show next that if $\kappa = \tau$ then $\dim \mathcal{N}(\beta) \geq n - 2p$ holds regardless the value of p. In that case we have that $\hat{B}_Y(V) = \hat{\mathcal{U}}(X)$ where $Y \in RE(\hat{\beta})$. From (5) we have $B_Y(N(X)) \subset \mathcal{U}(X)$. Then set $B_1 = B_Y|_{N(X)} \colon N(X) \to \mathcal{U}(X)$ and $N_1 = \ker B_1$. Hence $\dim N(X) \leq \dim N_1 + \tau$. From (5) and (7) we obtain

$$B_Z\eta = 0 \text{ if and only if } \langle\!\langle B_Z\eta, \hat{B}_Y(V)\rangle\!\rangle = 0$$

for $\eta \in N_1$ and $Z \in V^n$. Flatness of β and (7) give

$$\langle\!\langle B_Z\eta, \hat{B}_Y(V)\rangle\!\rangle = \langle\!\langle B_Z\eta, B_Y(V)\rangle\!\rangle = \langle\!\langle B_Z(V), B_Y\eta\rangle\!\rangle = 0$$

for $\eta \in N_1$ and any $Z \in V^n$. Thus $N_1 = \mathcal{N}(\beta)$, and we obtain using (8) that

$$\dim \mathcal{N}(\beta) = \dim N_1 \geq \dim N(X) - \tau \geq n - 2p.$$

By the above, it remains to consider the case $\kappa = 2$ and $\tau = 4$. If $\hat{B}_{Y_1}Z_1 = \hat{B}_{Y_2}Z_2$ then $S\hat{B}_{Y_1}Z_1 = S\hat{B}_{Y_2}Z_2$ for $Y_1, Y_2, Z_1, Z_2 \in V^n$ where S is given after (9). Then, by Lemma 6 the S invariant subspace $\hat{B}_{Y_1}(V) \cap \hat{B}_{Y_2}(V)$ is either even dimensional or zero. Hence, we can assume that there exist $Y_1, Y_2 \in RE^o(\beta) \cap RE(\hat{\beta})$ such that

$$\hat{\mathcal{U}}(X) = \hat{B}_{Y_1}(V) \oplus \hat{B}_{Y_2}(V).$$

We cannot have $B_{Y_j}(N(X)) = \mathcal{U}(X)$ for any $j = 1, 2$. If otherwise and since $p \leq 5$, then $\dim \mathcal{U}(Y_j) \leq p - 2 = 3$ in contradiction with $Y_j \in RE^o(\beta)$. Since the subspace $N(X)$ is J-invariant we have that if $(\eta, \bar{\eta}) \in B_{Y_j}(N(X))$ then also $(\bar{\eta}, -\eta) \in B_{Y_j}(N(X))$. Thus from Lemma 6 it remains to consider the case $\dim B_{Y_j}(N(X)) \leq 2$, $j = 1, 2$. Set $B_1 = B_{Y_1}|_{N(X)} \colon N(X) \to \mathcal{U}(X)$, $N_1 = \ker B_1$, $B_2 = B_{Y_2}|_{N_1} \colon N_1 \to \mathcal{U}(X)$ and $N_2 = \ker B_2$. By (5) and (7) we have

$$B_Z\eta = 0 \text{ if and only if } \langle\!\langle B_Z\eta, \hat{B}_{Y_j}(V)\rangle\!\rangle = 0,\ j = 1, 2,$$

for any $\eta \in N(X)$ and $Z \in V^n$. Flatness of β and (7) give

$$\langle\!\langle B_Z\eta, \hat{B}_{Y_j}(V)\rangle\!\rangle = \langle\!\langle B_Z\eta, B_{Y_j}(V)\rangle\!\rangle = \langle\!\langle B_Z(V), B_{Y_j}\eta\rangle\!\rangle = 0$$

for $\eta \in N_2$ and any $Z \in V^n$. Then $N_2 = \mathcal{N}(\beta)$ and (8) gives

$$\dim \mathcal{N}(\beta) \geq \dim N_2 \geq \dim N_1 - 2 \geq \dim N(X) - 4 \geq n - 2p,$$

and this concludes the proof. □

2. The proofs

The following application of Proposition 5 is the main ingredient in the proofs of the theorems in this paper.

Proposition 7. *Let V^n and U^p, $2p < n$ and $1 \leq p \leq 5$, be real vector spaces such that there is $J \in End(V)$ satisfying $J^2 = -I$ and U^p carries an either positive definite or Lorentzian inner product. Assume that the bilinear form $\beta \colon V^n \times V^n \to W^{p,p} = U^p \oplus U^p$ defined by (4) is flat with respect to the inner product (3). If $\dim \mathcal{N}(\beta) \leq n - 2p - 1$ then $\mathcal{U} = \mathcal{S}(\beta) \cap \mathcal{S}(\beta)^\perp$ satisfies that $\dim \mathcal{U} = s > 0$ is even. Moreover, if $L \subset U^p$ is the projection of $\mathcal{U}$ on the first factor of $W^{p,p}$ then we have:*

(i) *If the subspace L is nondegenerate then $\dim L = s$ and L inherits a positive definite inner product. With respect to the orthogonal splitting $U^p = L \oplus L^\perp$ we denote $\alpha_1 = \pi_L \circ \alpha$ and $\alpha_2 = \pi_{L^\perp} \circ \alpha$. Then*

$$\alpha_1(X, JY) = \alpha_1(JX, Y) \text{ for all } X, Y \in V^n$$

and

$$\dim \mathcal{N}(\alpha_2) \cap J\mathcal{N}(\alpha_2) \geq n - 2(p-s).$$

(ii) *If the subspace L is degenerate, let $0 \neq \delta \in L \cap L^\perp$. Then there is an orthogonal splitting $U^p = U_0 \oplus U_1^{s-2} \oplus U_2^{p-s}$, $s = 2$ or 4, so that $U_0 = span\{\delta, \zeta\}$ being $\zeta \in U^p$ a light-like vector where $\langle \delta, \zeta \rangle = 1$, $L = span\{\delta\} \oplus U_1^{s-2}$ and such that $\alpha_j = \pi_{U_j} \circ \alpha$, $j = 0, 1, 2$, satisfy*

$$\langle \alpha(X,Y), \delta \rangle = 0 \;\text{ and }\; \alpha_1(X, JY) = \alpha_1(JX, Y) \;\text{ for all }\; X, Y \in V^n$$

and

$$\dim \mathcal{N}(\alpha_2) \cap J\mathcal{N}(\alpha_2) \geq n - 2(p-s).$$

Proof. By Proposition 5 we have $s > 0$. If $0 \neq (\xi, \bar{\xi}) \in \mathcal{U}$, then

$$(\xi, \bar{\xi}) = \sum_i \beta(X_i, Y_i) = \sum_i (\alpha(X_i, Y_i), \alpha(X_i, JY_i))$$

and

$$0 = \langle\!\langle \beta(X,Y), (\xi, \bar{\xi}) \rangle\!\rangle = \langle \alpha(X,Y), \xi \rangle - \langle \alpha(X, JY), \bar{\xi} \rangle$$

for any $X, Y \in V^n$. Then $(\bar{\xi}, -\xi) = \sum_i \beta(X_i, JY_i) \in \mathcal{S}(\beta)$ and

$$\langle\!\langle (\beta(X,Y), (\bar{\xi}, -\xi) \rangle\!\rangle = \langle \alpha(X,Y), \bar{\xi} \rangle + \langle \alpha(X, JY), \xi \rangle = 0$$

for any $X, Y \in V^n$. Therefore also $(\bar{\xi}, -\xi) \in \mathcal{U}$. From Lemma 6 it follows that s is even and that

$$\pi_1(\mathcal{U}) = L = \pi_2(\mathcal{U}), \tag{10}$$

where $\pi_j \colon W^{p,p} \to U^p$, $j = 1, 2$, denote the projections onto the factors.

*C*ase (i): We have that the inner product induced on L is positive definite. If otherwise let $\delta \in L$ be time-like. By (10) there is $\bar{\delta} \in L$ such that $(\delta, \bar{\delta}) \in \mathcal{U}$. Then $0 = \langle (\delta, \bar{\delta}), (\delta, \bar{\delta}) \rangle = \langle \delta, \delta \rangle - \langle \bar{\delta}, \bar{\delta} \rangle$ and hence also $\bar{\delta}$ is time-like. Since $(\bar{\delta}, -\delta) \in \mathcal{U}$ as seen above and $0 = \langle\!\langle (\delta, \bar{\delta}), (\bar{\delta}, -\delta) \rangle\!\rangle = 2\langle \delta, \bar{\delta} \rangle$, we obtain a contradiction with the signature of U^p.

We have

$$\begin{aligned}
\beta(X,Y) &= (\alpha(X,Y), \alpha(X,JY)) = (\alpha_1(X,Y) + \alpha_2(X,Y), \alpha_1(X,JY) + \alpha_2(X,JY)) \\
&= (\alpha_1(X,Y), \alpha_1(X,JY)) + (\alpha_2(X,Y), \alpha_2(X,JY)) \\
&= \beta_1(X,Y) + \beta_2(X,Y).
\end{aligned}$$

Since $\mathcal{U}$ is an isotropic subspace then $\pi_1|_{\mathcal{U}} \colon \mathcal{U} \to L$ is an isomorphism and thus $\dim L = s$. Then by dimension reasons we have $\mathcal{U}^s = \mathcal{S}(\beta_1) \subset L^s \oplus L^s$. Hence β_1 is null and

$$0 = \langle\!\langle \beta_1(X,Y), \beta_1(Z,W) \rangle\!\rangle = \langle \alpha_1(X,Y), \alpha_1(Z,W) \rangle - \langle \alpha_1(X,JY), \alpha_1(Z,JW) \rangle.$$

Then $T \colon \mathcal{S}(\alpha_1) \to \mathcal{S}(\alpha_1)$ defined by $T\alpha_1(X,Y) = \alpha_1(X,JY)$ is a linear isometry and

$$\alpha_1(JX,Y)=\alpha_1(Y,JX)=T\alpha_1(Y,X)=T\alpha_1(X,Y)=\alpha_1(X,JY).$$

Being β flat and β_1 null, then also β_2 is flat. Since the subspace $\mathcal{S}(\beta_2)$ is nondegenerate we have from Proposition 5 that $\dim \mathcal{N}(\beta_2)\geq n-2(p-s)$. To conclude the proof of this case observe that $\mathcal{N}(\beta_2)=\mathcal{N}(\alpha_2)\cap J\mathcal{N}(\alpha_2)$.

Case (ii): Let $\bar{\delta}\in L$ be such that $(\delta,\bar{\delta}),(\bar{\delta},-\delta)\in\mathcal{U}$. Since the inner product on U^p has Lorentzian signature, then the vectors $\delta,\bar{\delta}$ must be linearly dependent. Thus $(\delta,0),(0,\delta)\in\mathcal{U}$, and hence

$$0=\langle\langle\beta(X,Y),(\delta,0)\rangle\rangle=\langle\alpha(X,Y),\delta\rangle$$

for all $X,Y\in V^n$.

Assume $s=2$, in which case $\mathcal{U}=\operatorname{span}\{(\delta,0),(0,\delta)\}$. Fix a light-like vector $\zeta\in U^p$ such that $\langle\zeta,\delta\rangle=1$. Set $U_0=\operatorname{span}\{\delta,\zeta\}$ and define U_2 by the orthogonal decomposition $U^p=U_0^2\oplus U_2^{p-2}$. We have $\alpha=\alpha_0+\alpha_2$, where

$$\alpha_0(X,Y)=\langle\alpha(X,Y),\zeta\rangle\delta.$$

Hence $\beta=\beta_0+\beta_2$, where

$$\beta_0(X,Y)=\langle\alpha(X,Y),\zeta\rangle(\delta,0)+\langle\alpha(X,JY),\zeta\rangle(0,\delta)$$

and

$$\beta_2(X,Y)=(\alpha_2(X,Y),\alpha_2(X,JY)). \tag{11}$$

Because β is flat and β_0 is null, then β_2 is flat. Since the subspace $\mathcal{S}(\beta_2)$ is nondegenerate, then Proposition 5 gives $\dim\mathcal{N}(\beta_2)\geq n-2p+4$.

Assume $s=4$. Then there are space-like vectors $\xi,\bar{\xi}\in L$ such that

$$\mathcal{U}=\operatorname{span}\{(\delta,0),(0,-\delta),(\xi,\bar{\xi}),(\bar{\xi},-\xi)\}.$$

Set $U_1=\operatorname{span}\{\xi,\bar{\xi}\}$ and fix light-like vector $\zeta\perp U_1$ such that $\langle\delta,\zeta\rangle=1$ is satisfied. Let $\beta_j\colon V^n\times V^n\to U_j\oplus U_j$ be given by

$$\beta_j(X,Y)=(\alpha_j(X,Y),\alpha_j(X,JY)),\ j=0,1,2$$

where $U_0=\operatorname{span}\{\delta,\zeta\}$ and the subspace U_2 is defined by the orthogonal decomposition $U^p=U_0\oplus U_1\oplus U_2$. Then $\beta=\beta_0+\beta_1+\beta_2$ where β_0 and β_1 are null.

If $T\colon U_1\to U_1$ be the linear isometry defined by $T\alpha_1(X,Y)=\alpha_1(X,JY)$, then

$$\alpha_1(JX,Y)=\alpha_1(Y,JX)=T\alpha_1(Y,X)=T\alpha_1(X,Y)=\alpha_1(X,JY).$$

Since β_2 is flat and $\mathcal{S}(\beta_2)$ is a nondegenerate subspace, then Proposition 5 gives that $\dim\mathcal{N}(\beta_2)\geq n-2p+8$, and this concludes the proof. $\square$

Let $f\colon M^{2n}\to\mathbb{R}^{2n+p}$ be a conformal immersion of a simply connected Kaehler manifold free of points where all sectional curvatures vanish and let $\alpha^F\colon TM\times TM\to N_FM$ be the second fundamental form of its isometric light-cone representative $F\colon M^{2n}\to\mathbb{V}^{2n+p+1}\subset\mathbb{L}^{2n+p+2}$. At any point $x\in M^{2n}$, let $\beta\colon T_xM\times T_xM\to W^{p+2,p+2}=N_FM(x)\oplus N_FM(x)$ be the bilinear form given by

$$\beta(X,Y) = (\alpha^F(X,Y), \alpha^F(X,JY)) \tag{12}$$

where the inner product in $W^{p+2,p+2}$ is as in (3). Using the Gauss equation and that the curvature tensor of M^{2n} satisfies $J \circ R(X,Y) = R(X,Y) \circ J$ it is easy to verify that β is flat. Moreover, since

$$\langle \alpha^F(X,Y), F\rangle = -\langle X, Y\rangle \tag{13}$$

we have that $\mathcal{N}(\beta) \subset \mathcal{N}(\alpha^F) = \{0\}$.

Proof of Theorem 1. We apply Proposition 7 at $x \in M^{2n}$ to $\beta\colon T_xM \times T_xM \to W^{3,3}$ defined by (12) in terms of α^F satisfying (13). Since $0 = \dim \mathcal{N}(\beta) \leq 2n-7$ then $s(x)$ is even. Being $\mathcal{U}$ an isotropic subspace of $W^{3,3}$ then $s(x) = 2$. We also have that L is a degenerate subspace. In fact, if otherwise, by part (i) there exists an orthogonal splitting $N_FM(x) = L \oplus L^\perp$ such that L inherits a positive definite inner product and $L^\perp = \operatorname{span}\{\eta\}$ where $\eta \in N_FM(x)$ is a unit time-like vector. Moreover, the J-invariant subspace $\Delta = \mathcal{N}(\beta_2)$ satisfies $\dim \Delta \geq 2n-2$ and, since $L = \operatorname{span}\{\xi, \bar{\xi}\}$ where $(\xi, \bar{\xi}) \in \mathcal{U}$, then the shape operators of F satisfy

$$A^F_{\bar{\xi}} = -J \circ A^F_{\xi} \ \text{ and } \ \Delta = \ker A^F_\eta.$$

Since $F \in N_FM(x)$ then $F = a\xi + b\bar{\xi} + c\eta$. Hence

$$JZ = -JA^F_F Z = -J(aA^F_\xi Z + bA^F_{\bar{\xi}} Z) = A^F_{b\xi - a\bar{\xi}} Z$$

for any $Z \in \Delta$. Therefore $J|_\Delta = A^F_{b\xi - a\bar{\xi}}|_\Delta$, and this is a contradiction.

Since the subspace L is degenerate, by part (ii) at any point there is a splitting $N_FM = \operatorname{span}\{\delta, \zeta\} \oplus U_2$ such that $A^F_\delta = 0$ and the J-invariant subspace $\Delta = \mathcal{N}(\beta_2)$ satisfies $\dim \Delta \geq 2n-2$. Moreover, since M^{2n} is free of points where all sectional curvatures vanish then we have $\dim \Delta = 2n-2$.

Because $F, \delta \in N_FM(x)$ are linearly independent then $\langle F, \delta\rangle \neq 0$. We take $\zeta = F$ and determine δ by requiring that $\langle \delta, F\rangle = 1$. Hence, we have a normal basis $\{\delta, F, \xi\}$ with $\xi \perp \operatorname{span}\{\delta, F\}$ of unit length such that

$$A^F_\delta = 0, \ \ A^F_F = -I \ \text{ and } \ \Delta = \ker A^F_\xi. \tag{14}$$

We have that $\mathcal{U} = \mathcal{S}(\beta) \cap \mathcal{S}(\beta)^\perp$ has constant dimension and hence it is smooth. It follows easily that also the frame $\{\delta, F, \xi\}$ can be taken to be smooth.

We claim that the vector field δ is parallel in the normal connection, and hence it is constant in the ambient space. The Codazzi equation gives

$$A^F_{\nabla^\perp_X \delta} Y = A^F_{\nabla^\perp_Y \delta} X$$

for any $X, Y \in TM$. Let $\theta\colon TM \to \operatorname{span}\{F, \xi\}$ be the linear map given by $\theta(X) = (\nabla^\perp_X \delta)_{\operatorname{span}\{F,\xi\}}$. Suppose that δ is not parallel, that is, that $\theta \neq 0$. It follows that $A_{\operatorname{Im}\theta} \ker\theta = 0$. But in view of (14) this is easily seen to be in contradiction with the assumption that M^{2n} is free of points where all sectional curvatures vanish. Now by Proposition 4 there is an isometric immersion $g\colon M^{2n} \to \mathbb{R}^{2n+1}$ so that $F = \Psi \circ g$. Being F the isometric light-cone representative of f as well as g, then we have by Proposition 3 that f and g are conformal immersions. □

For the proof of Theorem 2 we need the following two technical results.

Lemma 8. *Let $g\colon M^n \to \mathbb{R}^{n+1}$ be an isometric immersion and let $f\colon M^n \to \mathbb{R}^{n+p}$ be a conformal immersion. Then let $G\colon M^n \to \mathbb{V}^{n+2} \subset \mathbb{L}^{n+3}$ and $F\colon M^n \to \mathbb{V}^{n+p+1} \subset \mathbb{L}^{n+p+2}$ be the isometric light-cone representatives of g and f, respectively. Given an open subset $U \subset M^n$, there exists a conformal immersion $h\colon V \to \mathbb{R}^{n+p}$ of an open subset $V \supset g(U)$ of $\mathbb{R}^{n+1}$ such that $f|_U = h \circ g|_U$ if and only if there exists an isometric immersion $H\colon W \to \mathbb{V}^{n+p+1}$ of an open subset $W \subset \mathbb{V}^{n+2}$ with $G(U) \subset W$ such that $F|_U = H \circ G|_U$.*

Proof. See Proposition 2 in [2] or Proposition 2 in [8]. □

Lemma 9. *Let $F\colon M^n \to \mathbb{V}^{n+3} \subset \mathbb{L}^{n+4}$ be an isometric immersion and let ξ be a normal vector field of unit length that satisfies $\langle \xi, F\rangle = 0$, rank $A^F_\xi = 1$ and that is parallel along $\ker A^F_\xi$. Then, there exist open subsets $V \subset M^n$ and $W \subset \mathbb{V}^{n+2}$ and local isometric immersions $G\colon V \to \mathbb{V}^{n+2}$ and $H\colon W \to \mathbb{V}^{n+3}$ with $G(V) \subset W$ such that $F|_V = H \circ G$.*

Proof. See Lemma 2 in [2]. □

Proof of Theorem 2. We proceed by making use of the definitions and notations from the proof of Theorem 1. From (2) we have that $\mathcal{N}(\beta) = 0$. Proposition 7 applied to $\beta\colon T_xM \times T_xM \to W^{4,4}$ at $x \in M^{2n}$ gives $s(x) = 2$ or 4. In what follows we work on an open dense subset M_* of M^{2n} where $\dim \mathcal{S}(\beta)$ is locally constant. Let M_2 be the open subset of M_* where $s(x) = 2$. A similar argument as in the proof of Theorem 1 gives that the subspace L is degenerate at each point of M_2. Then along M_2 there is a smooth orthogonal splitting of the normal bundle of F as $N_FM = \operatorname{span}\{\delta, F\} \oplus P$ with $\langle \delta, F\rangle = 1$ such that $A^F_\delta = 0$, $A^F_F = -I$ and $\Delta = \mathcal{N}(\beta_2)$ satisfies $\dim \Delta \geq 2n - 4$ where β_2 is defined by (11).

Let $P = \operatorname{span}\{\xi_1, \xi_2\}$ be a local smooth orthonormal frame. In the sequel, we work on a connected component M_2' of the open subset of M_2 where $\dim \Delta$ and the ranks of the $A^F_{\xi_j}$'s are locally constant. The corresponding Codazzi equations are

$$\begin{aligned}&\nabla_X A^F_{\xi_i} Y - A^F_{\xi_i}\nabla_X Y + \langle \nabla^\perp_X \xi_i, \delta\rangle Y - \langle \nabla^\perp_X \xi_i, \xi_j\rangle A^F_{\xi_j} Y \\ &\quad = \nabla_Y A^F_{\xi_i} X - A^F_{\xi_i}\nabla_Y X + \langle \nabla^\perp_Y \xi_i, \delta\rangle X - \langle \nabla^\perp_Y \xi_i, \xi_j\rangle A^F_{\xi_j} X,\ 1 \leq i \neq j \leq 2,\end{aligned} \tag{15}$$

for any $X, Y \in \mathfrak{X}(M)$. It follows that

$$A^F_{\xi_j}[S,T] = \langle \nabla^\perp_S \xi_j, \delta\rangle T - \langle \nabla^\perp_T \xi_j, \delta\rangle S$$

for any $S, T \in \Gamma(\Delta)$. Hence $\langle \nabla^\perp_S \xi_j, \delta\rangle = 0$, $j = 1, 2$, for any $S \in \Gamma(\Delta)$. Then (15) yields

$$-A^F_{\xi_i}\nabla_X S + \langle \nabla^\perp_X \xi_i, \delta\rangle S = \nabla_S A^F_{\xi_i} X - A^F_{\xi_i}\nabla_S X - \langle \nabla^\perp_S \xi_i, \xi_j\rangle A^F_{\xi_i} X,\ 1 \leq i \neq j \leq 2,$$

for any $S \in \Gamma(\Delta)$ and $X \in \mathfrak{X}(M)$. In particular,

$$\langle \nabla^\perp_X \delta, \xi_j\rangle \langle S, T\rangle = \langle \nabla_S T, A^F_{\xi_j} X\rangle,\ j = 1, 2, \tag{16}$$

for any $X \in \mathfrak{X}(M)$ and $S, T \in \Gamma(\Delta)$. Thus

$$(\nabla_S T)_{\operatorname{Im} A^F_{\xi_j}} = \langle S, T\rangle Z_j,\ j = 1, 2, \tag{17}$$

where $Z_j \in \Gamma(\operatorname{Im} A^F_{\xi_j})$ and $S, T \in \Gamma(\Delta)$. Now (16) reads as

$$\langle \nabla^\perp_X \delta, \xi_j\rangle = \langle A^F_{\xi_j} Z_j, X\rangle,\ j = 1, 2, \tag{18}$$

for any $X \in \mathfrak{X}(M)$.

We denote

$$R(x) = \text{span}\,\{A^F_{\xi_j} Z_j(x) : x \in M_2',\ j = 1, 2\}.$$

We claim that the open subset $N_2 \subset M_2'$ defined by $N_2 = \{x \in M_2' : \dim R(x) = 2\}$ is empty. In fact, the Codazzi equation for A^F_δ is

$$\langle \nabla^\perp_X \delta, \xi_1 \rangle A^F_{\xi_1} Y + \langle \nabla^\perp_X \delta, \xi_2 \rangle A^F_{\xi_2} Y = \langle \nabla^\perp_Y \delta, \xi_1 \rangle A^F_{\xi_1} X + \langle \nabla^\perp_Y \delta, \xi_2 \rangle A^F_{\xi_2} X \tag{19}$$

for any $X, Y \in \mathfrak{X}(M)$. We have from (18) that δ is parallel along $R^\perp$. Hence

$$\langle \nabla^\perp_Y \delta, \xi_1 \rangle A^F_{\xi_1} X + \langle \nabla^\perp_Y \delta, \xi_2 \rangle A^F_{\xi_2} X = 0 \tag{20}$$

for any $X \in \Gamma(R^\perp)$ and $Y \in \mathfrak{X}(M)$. In particular, the vectors $A^F_{\xi_1} X, A^F_{\xi_2} X$ cannot be linearly independent for any $X \in R^\perp$. If otherwise (20) yields that δ is parallel, and then (18) gives $R = 0$, a contradiction.

We argue that

$$R^\perp \subset \ker A^F_{\xi_1} \cap \ker A^F_{\xi_2}. \tag{21}$$

Suppose that $X \notin \ker A^F_{\xi_1} \cap \ker A^F_{\xi_2}$ for $X \in \Gamma(R^\perp)$. By the above $A^F_{\xi_1} X = \gamma A^F_{\xi_2} X$ where $A^F_{\xi_2} X \neq 0$ but $\gamma \in C^\infty(N_2)$ may vanish. We obtain from (20) that

$$\gamma \langle \nabla^\perp_Y \delta, \xi_1 \rangle + \langle \nabla^\perp_Y \delta, \xi_2 \rangle = 0$$

for any $Y \in \mathfrak{X}(M)$. Then (18) gives $\gamma A^F_{\xi_1} Z_1 + A^F_{\xi_2} Z_2 = 0$, which is a contradiction.

Since M^{2n} is free of points where all sectional curvatures vanish, we have from (21) that we may choose the frame $\{\xi_1, \xi_2\}$ such that $\ker A^F_{\xi_1} = R^\perp = \ker A^F_{\xi_2}$. From (17) we obtain $Z_1 = Z_2 = Z \in R$. Then (18) and (19) give

$$\langle A^F_{\xi_1} Z, X \rangle A^F_{\xi_1} Y + \langle A^F_{\xi_2} Z, X \rangle A^F_{\xi_2} Y = \langle A^F_{\xi_1} Z, Y \rangle A^F_{\xi_1} X + \langle A^F_{\xi_2} Z, Y \rangle A^F_{\xi_2} X \tag{22}$$

for any $X, Y \in \mathfrak{X}(M)$. By the Gauss equation (22) is equivalent to $R(X, Y)Z = 0$, and this is a contradiction since M^{2n} is free of points where all sectional curvatures vanish. Thus the claim that N_2 is empty has been proved.

Let $N_1 \subset M_2'$ be the open subset defined as $N_1 = \{x \in M_2' : \dim R(x) = 1\}$. Similarly as above, we obtain that δ is parallel along the hyperplane $R^\perp$ and that the vectors $A^F_{\xi_1} X, A^F_{\xi_2} X$ cannot be linearly independent for any $X \in R^\perp$. And from (18) we have that δ is not a parallel vector field. Hence, we can choose the frame $\{\xi_1, \xi_2\}$ for P such that

$$\langle \nabla^\perp_X \delta, \xi_2 \rangle = 0 \tag{23}$$

for any $X \in \mathfrak{X}(M)$. It now follows from (19) that $R^\perp \subset \ker A^F_{\xi_1}$.

We have $R^\perp = \ker A^F_{\xi_1}$. In fact, otherwise $A^F_{\xi_1} = 0$ and the Codazzi equation gives

$$\langle \nabla^\perp_X \xi_1, \delta \rangle S = -\langle \nabla^\perp_S \xi_1, \xi_2 \rangle A^F_{\xi_2} X$$

for any $S \in \Gamma(\Delta)$ and $X \in \mathfrak{X}(M)$. It follows that δ is a parallel vector field, and this is a contradiction.

We obtain from (18) and (23) that $A^F_{\xi_2} Z_2 = 0$. Since $A^F_{\xi_2}|_{\mathrm{Im}\, A^F_{\xi_2}}$ is an isomorphism, then $Z_2 = 0$. If $\mathrm{Im}\, A^F_{\xi_1} \subset \mathrm{Im}\, A^F_{\xi_2}$, it follows from (17) that $Z_1 = 0$. Thus $\mathrm{Im}\, A^F_{\xi_1} \not\subset \mathrm{Im}\, A^F_{\xi_2}$ and since M^{2n} is free of points where all sectional curvatures vanish we have that $2 \leq \dim \operatorname{Im} A^F_{\xi_2} \leq 3$. In fact, it holds that $\dim \operatorname{Im} A^F_{\xi_2} = 2$ since, otherwise, $\Delta^\perp = \operatorname{Im} A^F_{\xi_1} \oplus \operatorname{Im} A^F_{\xi_2}$. Then (17) gives $(\nabla_S T)_{\Delta^\perp} = \langle S, T\rangle Z_1$ for any $S, T \in \Delta$, and hence

$$\langle S, T\rangle JZ_1 = J(\nabla_S T)_{\Delta^\perp} = (J\nabla_S T)_{\Delta^\perp} = (\nabla_S JT)_{\Delta^\perp} = \langle S, JT\rangle Z_1.$$

Thus $Z_1 = 0$, and this is a contradiction.

If $Y, Z \in \Gamma(\ker A^F_{\xi_1})$ are linearly independent, then the Codazzi equation for $A^F_{\xi_1}$ is

$$A^F_{\xi_1}[Z, Y] = A^F_{\xi_2}(\langle \nabla^\perp_Y \xi_1, \xi_2\rangle Z - \langle \nabla^\perp_Z \xi_1, \xi_2\rangle Y).$$

Since $\operatorname{Im} A^F_{\xi_1} \not\subset \operatorname{Im} A^F_{\xi_2}$ and $A^F_{\xi_2}|_{\operatorname{Im} A^F_{\xi_2}}$ is an isomorphism, then $\langle \nabla^\perp_Y \xi_1, \xi_2\rangle = 0$ for any $Y \in \Gamma(\ker A^F_{\xi_1})$. Thus ξ_1 is parallel along $\ker A^F_{\xi_1}$.

By Lemma 9 there exist a simply connected open neighborhood $V \subset N_1$ of any $x \in N_1$, an open subset $W \subset \mathbb{V}^{2n+2}$ and local isometric immersions $G\colon V \to \mathbb{V}^{2n+2}$ and $H\colon W \to \mathbb{V}^{2n+3}$ with $G(V) \subset W$ such that $F|_V = H \circ G$. An elementary argument gives that there exists a conformal immersion $g\colon V \to \mathbb{R}^{2n+1}$ that has G as isometric light-cone representative; see pg. 7 of [10] or Proposition 9.9 of [7]. By Lemma 8 there exists a conformal immersion $h\colon U \to \mathbb{R}^{2n+2}$ such that $f|_V = h \circ g|_V$ with $g(V) \subset U$. Finally, by Theorem 1 applied to g we are as in part (ii) of Theorem 2.

Let N_0' be an open simply connected subset of the set $N_0 = \operatorname{int}\{x \in M_2 : R(x) = 0\}$. Then δ is constant on N_0' from (18). By Proposition 4, there is an isometric immersion $g_0\colon N_0' \to \mathbb{R}^{2n+2}$ whose isometric light-cone representative is $F|_{N_0'}$. From Proposition 3 we are in part (i) of Theorem 2.

Finally, let $M_4 \subset M^{2n}$ be the interior of the set $\{x \in M^{2n} : s(x) = 4\}$. Then $L(x)$ for $x \in M_4$ is a degenerate subspace since, otherwise, $N_F M_4(x) = L(x)$ which contradicts the fact that $L(x)$ has a positive definite inner product. By Proposition 7, there is a smooth orthogonal vector bundle decomposition $N_F M_4 = \operatorname{span}\{\delta, F\} \oplus U_1^2$ such that $A^F_\delta = 0$, $A^F_F = -I$ and $\langle \delta, F\rangle = 1$. Moreover, we have $U_1^2 = \operatorname{span}\{\xi_1, \xi_2\}$ where the smooth frame is orthonormal and $A^F_{\xi_1} = J \circ A^F_{\xi_2}$. Then the Codazzi equation for ξ_1 gives

$$JA_{\nabla^\perp_X \xi_2} Y - A_{\nabla^\perp_X \xi_1} Y = JA_{\nabla^\perp_Y \xi_2} X - A_{\nabla^\perp_Y \xi_1} X$$

for any $X, Y \in \mathfrak{X}(M)$. Expanding the equation we obtain

$$\begin{aligned} -\langle \nabla^\perp_X \xi_2, \delta\rangle JY + \langle \nabla^\perp_X \xi_2, \xi_1\rangle JA^F_{\xi_1} Y + \langle \nabla^\perp_X \xi_1, \delta\rangle Y - \langle \nabla^\perp_X \xi_1, \xi_2\rangle A^F_{\xi_2} Y \\ = -\langle \nabla^\perp_Y \xi_2, \delta\rangle JX + \langle \nabla^\perp_Y \xi_1, \xi_1\rangle JA^F_{\xi_2} X + \langle \nabla^\perp_Y \xi_1, \delta\rangle X - \langle \nabla^\perp_Y \xi_1, \xi_2\rangle A^F_{\xi_2} X \end{aligned}$$

for any $X, Y \in \mathfrak{X}(M)$. Then using that $A^F_{\xi_1} = J \circ A^F_{\xi_2}$ it follows that $\delta \in \Gamma(N_F M_4)$ is parallel, and hence constant in $\mathbb{L}^{n+4}$. Finally, we combine Propositions 3 and 4 to again conclude that we are in part (i) of Theorem 2. $\square$

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